Diagnosing Human-to-Robot Transfer Failures with a Transfer Gap Budget in Simulation

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Abstract: A failed robot rollout does not identify which stage of human-to-robot transfer failed. We introduce the *Transfer Gap Budget*, which replaces one pipeline component at a time with a declared oracle and measures the paired change in task success. A joint oracle measures non-additivity. In an injected-error study with 500 fresh seeds per arm, perception has the largest budget on both the UR5e and UR10e. A look-again correction, selected on development seeds, raises success by 26.8 and 25.8 pp, respectively, the largest observed gains among four corrections. On six demonstrations from three people, diagnosis and a revised retreat raise paired success from 0–35/60 to 25–60/60; these exploratory evaluations reuse seeds from earlier development. Simulated depth perception meets pre-declared success and path-error criteria on three manipulators. A separate G1 humanoid study finds 11/40 versus 31/40 placements for the same closed-loop controller across two runtime setups, with the cause unresolved. All robot evaluations are in simulation.

Keywords: human-to-robot transfer, failure diagnosis, oracle substitution

1 Introduction

Human-to-robot transfer requires motion estimation, retargeting, calibration, and executable timing. Aggregate success gives little guidance about which component to revisit, particularly with diverse human data [1]. H2RBench measures transfer and task progress [2]; we ask how much success correcting a pipeline component could recover.

The *Transfer Gap Budget* measures each component's net oracle gain. The gains can overlap. The largest development-set gain selects a correction for fresh evaluation. We test this protocol on two arms, examine real-video and depth-camera settings, and report humanoid runtime sensitivity.

2 Related Work

Human-to-robot transfer. H2RBench reports sim–real agreement and quantitative transfer gains by execution stage on a Franka arm [2]. We intervene on pipeline components. Human-as-Humanoid and HumanEgo use human video for robot learning [3, 4]. Hand-pose quality and motion mismatch affect cotraining [5]; the embodiment-gap survey describes the resulting adaptation work [6]. **Diagnosis and correction.** AHA, RoboFAC, and Guardian diagnose manipulation failures [7, 8, 9]. RHO and GaP search policy code and skill graphs [10, 11]. We instead use oracle gains to select from a fixed correction menu.

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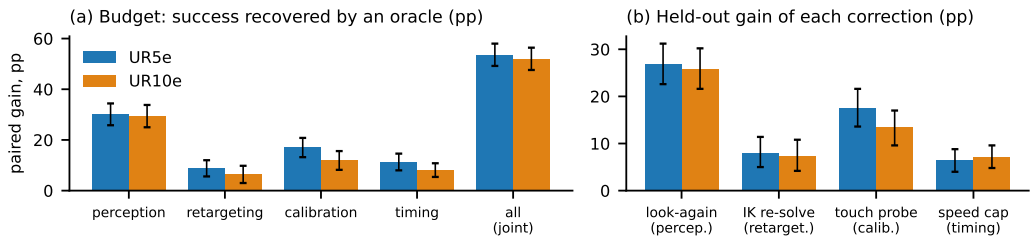


Figure 1: **Controlled-error study: 500 fresh seeds per arm.** (a) Paired success gains from single and joint oracles. (b) Gains from the four corrections; look-again was selected on development seeds. Error bars show paired bootstrap 95% intervals.

3 Method: the Transfer Gap Budget

Paired interventions. Let $\mathcal{K}$ contain perception, retargeting, tool-frame calibration, and execution timing. In the controlled-error study, *full* includes all four perturbations; *oracle*$_k$ removes only perturbation k, and *oracle-all* removes all four. Their ideal outputs are exact object pose, unperturbed IK, zero tool offset, and nominal timing. Each random stream is keyed by seed and component, preserving initial scenes and unreplaced error draws across conditions. For binary success $s_c(i)$ on seed i in condition c,

$$B_k = \frac{1}{n}\sum_{i=1}^{n}\big(s_{\text{oracle}_k}(i) - s_{\text{full}}(i)\big), \qquad \Delta_{\text{int}} = \sum_{k\in\mathcal{K}} B_k - B_{\text{all}}. \tag{1}$$

Here B_{all} uses the joint oracle in the same paired difference. Budgets are signed net gains: an oracle can both rescue and lose trials. The residual measures whether summed single-stage gains exceed (> 0) or fall below (< 0) the joint gain. It does not uniquely allocate failures or identify individual interactions.

Correction selection. The menu pairs perception with look-again, retargeting with an IK re-solve, calibration with a modelled touch probe, and timing with a speed cap. We select $k^\star = \arg\max_k B_k$ on development seeds, breaking ties in pipeline order. The driver requires a committed protocol and matching error constants before evaluation. Look-again and probing reduce assumed residual errors; their manoeuvres are not simulated. Outside oracle conditions, the controlled-error arm controller accesses object pose through the configured observation function, checked by unit tests.

Statistics. Wilson intervals describe individual success rates; paired percentile bootstrap intervals describe gains and the residual (95%; 5000 resamples in Section 5.1, 2000 for demos). Resampling preserves all condition scores for each seed. Demo intervals are descriptive because of adaptive seed reuse. Gains use percentage points (pp).

4 Experimental Setup

Task. MuJoCo simulates a fixed-base UR5e or UR10e with a Robotiq 2F-85 from MuJoCo Menagerie [12], a 4 cm block with ± 3 cm position jitter, and a 14 cm tray. A scripted IK controller uses simulated contact in a UR5e-tuned layout. Success requires a lift of at least 5 cm, release within 3 cm of the goal, and speed below 0.02 m/s for 0.5 s after release.

A: controlled errors. The injected magnitudes are assumptions, not estimates from human video. Both arms receive the same seed-dependent draws: Gaussian xy perception error ($\sigma = 1.5$ cm), joint offsets ($\sigma = 0.5^\circ$ per joint), a uniformly directed tool-frame offset with magnitude $\mathcal{U}(0, 3)$ cm, and a timing speed-up of $\mathcal{U}(1.0, 4.5)$ relative to robot pace. The corrections reduce perception noise to 0.4 cm, joint-offset noise to 0.05°, or calibration noise to 2 mm per axis, or cap speed-up at $3\times$. Development seeds 0–59 select look-again on both arms. After freezing the protocol and four hypotheses, we evaluate ten conditions on 500 fresh seeds (3000–3499) per arm: *full*, four single oracles, the joint oracle, and four corrections applied to *full*.

Table 1: Successes on 60 paired evaluation seeds: uncorrected → revised correction stack. P1–P3 denote demonstrators; P2 is evaluated with two stacks. Seeds were reused after development.

Demonstration	UR5e	UR10e
P1 notebook	0→ 57	0→ 60
P1 markers	0→ 51	0→ 25
P1 orange marker	0→ 46	0→ 58
P1 notepad	0→ 58	0→ 58
P2 box (own stack)	11→ 60	4→ 55
P2 box (P1 stack)	11→ 60	4→ 50
P3 recording (P1 stack)	35→ 59	30→ 56

B: real-video motion. Five pick-and-place cycles from two people in two freely licensed videos, plus one cycle from a third person's 50 s recording, supply motion and timing. GEM-X [13] estimates hand trajectories; grasp and release frames are manually annotated. Reach and retreat are anchored to the simulated grasp and release points; the carry path is rotated and scaled in the table plane to connect them. The video scene is not reconstructed. Perception and calibration errors remain injected; no object detector is used. Without a clean hand-motion reference, retargeting is excluded from the real-video oracle analysis. Each demo–arm pair uses 60 evaluation seeds. We report mean leg-wise path RMSE against the retargeted reference, after normalising time within each leg.

C: depth perception. A geometric estimator fits the table plane, clusters points above it, and selects a cube and tray using shape priors. Estimated position, yaw, and size drive the controller. Camera intrinsics and pose are known in nominal conditions; stress tests inject calibration errors. We test 5, 6, and 7 cm cubes on the two arms and a third manipulator (UR10e with a Robotiq 2F-140), with 40 seeds per size: 120 episodes per condition. **D: humanoid.** A Unitree G1 with Dex3 hands uses SONIC whole-body control [14]. We test human-path and IK-planned controllers, scoring touch, grasp (5 cm lift with contact), and placement within 5 cm of the goal. We do not estimate all four stage budgets here.

5 Results

5.1 Controlled errors on two arms

Uncorrected success is 232/500 on the UR5e and 240/500 on the UR10e. Removing all errors recovers 53.6 and 52.0 pp. Perception has the largest budget: 30.0 [25.8, 34.4] pp on the UR5e and 29.4 [25.0, 33.8] pp on the UR10e. Calibration follows (17.0 and 12.0 pp), then timing (11.2 and 8.0 pp) and retargeting (8.8 and 6.4 pp); all single-stage intervals exclude zero (Fig. 1a).

The fresh sample gives positive retargeting budgets on both arms. The interaction residual is 13.4 [6.2, 20.8] pp on the UR5e and 3.8 [-2.6, 10.4] pp on the UR10e. The pilot's sign reversal is unsupported; a direct arm contrast was not tested.

Look-again, selected from the development trials, raises success from 232 to 366/500 on the UR5e and from 240 to 369/500 on the UR10e. The paired gains are 26.8 [22.6, 31.2] pp and 25.8 [21.6, 30.2] pp, respectively. These are the largest observed gains. Calibration probing gives 17.6 and 13.4 pp; IK re-solving and the speed cap each give 6–8 pp (Fig. 1b). Pairwise correction differences were not tested.

5.2 Real-video demonstrations

The notebook demo fails uncorrected on both arms at about $6\times$ robot pace. Sequential diagnosis selects a $3\times$ speed cap, look-again, calibration probing, then a tighter $2\times$ cap. With the revised retreat, the stack reaches 57/60 and 60/60 (Table 1). RMSE to the retargeted reference falls from 13.9 to 4.9 cm on the UR5e and from 9.1 to 2.9 cm on the UR10e, at the cost of slowing the motion.

The original stack scored 0/60 on all 3 additional P1 cycles on both arms. Their copied retreat paths intersect the tray after release, and failures persist with all available oracles. On P2, the earlier P1 stack reduced UR5e success from 11/60 to 0/60. We therefore added a 5 cm lift before retreating along the human path, selected on development seeds 0–19. The revised stack improves all table cells; paired descriptive intervals exclude zero. The added lift increases path RMSE by 2.7–4.1 cm relative to the earlier stack. The revised P1 stack transfers to P2 and P3 without retuning; P1 markers remain difficult on the UR10e (25/60). These are exploratory results: seeds 1000–1059 were used before the revision.

5.3 Depth perception in the loop

All 30/30 pre-declared acceptance cells reach the 90% point-estimate threshold (113–120 successes per 120 episodes). They cover yaw-aware estimation, distractors, 10 mm depth noise, arm occlusion with a retry, and combined stressors on three manipulators. Report-only stress tests expose limits: a 2°/20 mm calibration error leaves the UR5e at 25/120, and arm occlusion without retry gives 0–16/120 across robots. Ignoring yaw yields 101/120 and 107/120 on the 2F-85 arms even with exact object position.

For the notebook path at robot pace, depth versus ground-truth success is 109 versus 108/120 (UR5e), 112 versus 111/120 (UR10e), and 120 versus 118/120 (third arm). Depth is 0.8–1.7 pp higher in these trials; path RMSE differs by less than 0.01 cm. All three meet the point-estimate margins of 10 pp success loss and 0.5 cm added RMSE; no inferential non-inferiority test was performed. Under combined depth stress, success falls to 88/120, 88/120, and 91/120, respectively.

5.4 G1 humanoid: timing and runtime sensitivity

Human-path controller. With oracle targets (ten seeds, three rollouts each), human pace touches but never grasps the block in either setup. At a $3\times$ cap, (grasps, placements) are (18, 8) on a GPU VM and (15, 2) in the container; at robot pace, (24, 20) and (24, 14). All counts are out of 30. Slowing helps both setups; the container does not improve placements.

Closed-loop controller. With IK-planned targets, the same frozen controller places 11/40 objects in a volume-based installation ("host") and 31/40 in a packaged container. Both run in cluster pods on the same GPU model and seeds 1200–1239. Discordant placements favour the container 20 to 0 (two-sided exact McNemar $p = 1.9 \times 10^{-6}$). Code and environment checks did not identify the cause. Five runs of the same ten seeds give 15/50 lifts of at least 5 cm on the host and 46/50 across container variants (repeated attempts, not 50 independent seeds). Host counts range from 1 to 7 of ten. In a separate four-seed check, an image rebuild with MuJoCo 3.15.0 gives 0/4 lifts versus 4/4 after restoring 3.14.0. The `filelock` version also changed, so this comparison does not isolate MuJoCo. Both setups in the 40-seed study use 3.14.0. The cause remains unknown.

6 Limitations and Conclusion

The study uses one simulated task, scripted arm controllers, a UR5e-tuned layout, and assumed error and correction magnitudes. Videos supply translation and timing, with manual keyframes. Six demonstrations, unreconstructed scenes, and adaptive seed reuse limit generalisation. Look-again and probing incur no simulated time cost. Choosing look-again on both arms does not establish an advantage over always selecting it. The G1 discrepancy remains unexplained.

Perception has the largest budget under the declared model; its selected correction gives the largest observed gain on fresh seeds. Video trials expose infeasible retreat paths. This supports diagnosis in simulation; broader transfer and real-robot performance remain untested.

An AI assistant helped draft parts of this manuscript and code; reported values were checked against saved results.

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